

Infineon Linux AIROC™ Bluetooth® release notes

About this document

Scope and purpose

This document offers a summary of the Bluetooth® firmware release, highlighting known bug fixes and updated specifications for AIROC™ combo chip (CYW55513/2/1, CYW55573/2/1, CYW43022, CYW43439, CYW43455 and CYW4373) in the latest Linux release. It also includes information on the proprietary AIROC™ Bluetooth® Stack (BTSTACK) and related code examples.

Intended audience

This document is intended for anyone who uses the AIROC™ Wi-Fi + Bluetooth® combo chips on a Linux host.

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Overview

1 Overview

1.1 High-level summary

This document outlines the new features and bug fixes for CYW55513/2/1, CYW55310, CYW55573/2/1, CYW43022, CYW43439, CYW43455, and CYW4373 in the Bluetooth® Linux software release.

This software release includes the following updates:

- Compliance with the latest Bluetooth® specifications
- Bug fixes
- AIROC™ Bluetooth® Stack updates
- Code example updates

1.2 Release updates

- Latest AIROC™ Bluetooth® Stack v5.0.0
- Release of new code examples to the SDK package:
 - Linux Bluetooth® LE Audio Player with latest BTSTACK support
 - Linux Bluetooth® LE Audio Headset with latest BTSTACK support
- Updates to controller firmware

1.3 Device firmware revision details

Table 1 Bluetooth® firmware revision details

Device	Bluetooth® firmware version	Bluetooth® version	QDID
CYW55513/2/1 & CYW55310	001.002.032.0253.0000	6.0	Q338655
CYW55573/2/1	001.002.087.0415.0000	6.0	Q345126
CYW43022	003.002.024.0044.0000	5.4	Q366491
CYW43439	001.003.016.0076.0000	5.4	Q370559
CYW43455	003.001.025.0196.0000	5.3	183141
CYW4373	001.001.025.0138.0000	5.4	Q370784

Note: AIROC™ Bluetooth® Stack BTSTACK v5.0.0 is validated on the Linux kernel v 6.12 with Raspberry Pi CM4.

Firmware changes

2 Firmware changes

This section provides details on the firmware updates, including bug fixes and enhancements, which may affect various AIROC™ devices.

2.1 CYW55513/2/1

CYW55513, CYW55512, and CYW55511 are designed to address the requirements of the Internet of Things (IoT) devices that require minimal power consumption and compact size.

- CYW55513/CYW55512/CYW55511 support Bluetooth®/Bluetooth® Low Energy
- Bluetooth® 6.0 certified firmware
- Enhancement of the isochronous adaptation layer (ISOAL), a Bluetooth® 6.0 feature
- Support Bluetooth® Low Energy (LE) with LE-audio
 - LE- Audio with LC3 Codec on the device (Offload mode)
 - LE- Audio with LC3 Codec on the host (HCI mode)
- LE isochronous channels
- Bluetooth® 5.0/5.1/5.2/5.3/5.4/6.0 features:
 - LE long range (LR)
 - LE 2 Mbps, LE 1 Mbps
 - Advertising extensions

2.1.1 Bug fixes

- Improved LE- audio quality
- Improved BR/EDR (basic rate/enhanced data rate) audio

Firmware changes

2.1.2 Known issues

Table 2 Known issues for CYW55513/2/1

Supported features	Known issues
Coding scheme selection for advertising (CSSA)	None
Periodic advertising ADI support, LE channel classification, and LE enhanced connection update	LE CIS connection establishment fails when the corresponding LE-ACL connection is subrated.
LE core Isochronous channels, LE power control	Disconnection due to MIC failure is observed in certain scenarios with IUT as peripheral in a single unicast LE-audio stream. Disconnection due to MIC failure is observed in certain scenarios with IUT as the central and audio sink in a single unicast LE-audio stream. Disconnection due to connection timeout is observed in certain scenarios with the IUT as a peripheral in a single unicast LE- audio stream. Auracast synchronizer intermittently fails to establish sync to the Auracast broadcaster.
Periodic advertising sync transfer	None
Extended advertising feature, 2 Mbps LE	None
LE privacy, data packet length extension (DPLE)	None

2.2 CYW55573/2/1

2.2.1 Bug fixes

- Bug fixes and improvements in LE ISOC

2.2.2 Known issues

- Bluetooth® ACL disconnection occurs when using EV3 on eSCO and 2/3 Mbps packet on Bluetooth® ACL, as IUT does not respond to 2/3 Mbps data packets on Bluetooth® ACL
- The Auracast synchronizer intermittently fails to establish sync to the Auracast broadcaster
- Disconnection due to MIC failure occurs in certain scenarios with IUT as central and audio sink in a single unicast LE-audio streamCYW43022

2.3 CYW43022

2.3.1 Bug fixes

- Updated qualification to the TCRL 2025-2

2.3.2 Known issues

- No issues reported during internal sanity testing

Firmware changes

2.4 CYW43439

2.4.1 Bug fixes

- Updated qualification to the TCRL pkg100 certification benchmark

2.4.2 Known issues

- No issues reported during internal sanity testing

2.5 CYW43455

2.5.1 Bug fixes

- Bluetooth® 5.3 updates

2.5.2 Known issues

- No issues reported during internal sanity testing

2.6 CYW4373

2.6.1 Bug fixes

- Updated qualification to the TCRL pkg100 certification benchmark

2.6.2 Known issues

- No issues reported during internal sanity testing

2.7 BTSTACK release – v5.0.0

- HFP v1.9 capability with super wide band (LC3) in SCO
- Supports A2DP 1.4
- Major updates for extended advertising and scanning APIs
- Major updates for ISOC APIs
- Updated parameter list for `wiced_ble_isoc_remove_data_path` API
- Deprecated API `wiced_bt_dev_delete_bonded_device`. The stack does not store any bonding information. For applications that use Bluetooth® LE mode, invoke `wiced_bt_dev_remove_device_from_address_resolution_db` to remove the device from the resolving list
- For detailed changelogs, refer to the [BTSTACK changelogs](#)

Bluetooth® Linux code examples and supported chips

3 Bluetooth® Linux code examples and supported chips

Table 3 Bluetooth® Linux code examples and supported chips

Code example	Feature demonstration	Supported chips
Linux Bluetooth® Find me	Demonstrates the Find me profile, which defines the behavior when a button is pressed on one device to cause an alerting signal on a peer device.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571, CYW43022, CYW43439, CYW4373, CYW43455
Linux Bluetooth® hello sensor	Demonstrates GATT database and device configuration initialisation, sending data to the client and processing write requests from the client	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571, CYW43022, CYW43439, CYW4373, CYW43455
Linux Bluetooth® Wi-Fi onboarding	Demonstrates a feature that enables devices to connect to a Wi-Fi access point without requiring a physical interface. This eliminates the need for cumbersome cables and allows for communication of Wi-Fi information through Bluetooth®, including SSID, password, control command, and device Wi-Fi status, providing a true wireless solution.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571, CYW43022
Linux Bluetooth® headset	Multiple profile CE, which demonstrates the use cases and abilities of audio-related functions like A2DP, AVRCP, CT, and HFP. It also supports GFPS, making it easier for the user to pair devices with as little user interaction as possible.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571, CYW43022, CYW43439, CYW4373, CYW43455
Linux Bluetooth® wake on LE	Demonstrates use of the APCF function to achieve Wake on LE.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571, CYW43022, CYW43439, CYW4373, CYW43455
Linux Bluetooth® LE-audio player	Demonstrates the implementation of the Basic audio profile (role: Unicast source / Broadcast source).	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571
Linux Bluetooth® LE-audio headset	Demonstrates an implementation of the Basic audio profile (role: Unicast sink / Broadcast sink).	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571
Linux Bluetooth® serial port profile	Demonstrate Bluetooth® SPP server using AIROC™ Wi-Fi + Bluetooth® combo chip.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571, CYW43022, CYW43439, CYW43455, CYW4373

Bluetooth® Linux code examples and supported chips

3.1 Bluetooth® libraries

Table 4 Bluetooth® libraries

Libraries and middleware	Library details
bluetooth-linux	The bluetooth-linux is the adaptation layer (porting layer) between the Linux Bluetooth® application code example and Infineon's BTSTACK running on the Linux-based platforms. The porting layer provides AIROC™ BTSTACK initialization and implements platform interfaces to provide OS and memory services, enabling communication between the BTSTACK and the Bluetooth® controller.
bt-audio-profiles	Contains source and header files for A2DP, AVRCP, HFP, and SPP profiles.
btsdk-gfps	Contains source and header files for the Google fast pairing service.
btstack	BTSTACK is Infineon's Bluetooth® Host Protocol Stack implementation. The stack is optimised to work with Infineon Bluetooth® controllers. The BTSTACK supports Bluetooth® BR/EDR and Bluetooth® LE core protocols.
le-audio-profiles-linux	Provides implementation of various LE Audio profiles, a GATT interface utility for LE audio code example, an audio module library, and an ISOC data handler interface.
fw	Folder contains various firmware files for the AIROC™ combo chip (CYW55513/2/1, CYW55573/2/1, CYW43022, CYW43439, CYW43455 and CYW4373).

Tools and documentation

4 Tools and documentation

See the [Table 5](#) for more details.

Table 5 Tools and documentation

Tools	Scope
Manufacturing Bluetooth® Test tool (MBT)	The manufacturing Bluetooth® Test tool (MBT) is used to test and verify the RF performance of the Infineon Bluetooth® Classic and Bluetooth® Low Energy devices on Linux platforms. See the MBT on GitHub .
AIROC™ Bluetooth® Test and Debug tool	AIROC™ Bluetooth® Test and Debug tool is a GUI tool for testing and debugging Infineon Bluetooth® devices. AIROC™ Bluetooth® Test and Debug Tool connects to the Bluetooth® devices at the HCI protocol layer and currently supports HCI UART and HCI USB transport interfaces. The tool allows users to send Bluetooth® HCI commands and receive Bluetooth® HCI events from the Bluetooth® controller of the connected devices. Download the tool from the AIROC™ Bluetooth® Test and Debug webpage.

Glossary

Glossary

A2DP

advanced audio distribution profile

ACL

asynchronous connection-oriented

ADI

Advertisement data information

AEC-CCM

advanced encryption standard-counter with cipher block chaining message authentication code

API

application programming interface

APCF

advertising packet content filter

AVRCP

audio/video remote control protocol

BIS

broadcast isochronous stream

BR

basic rate

BTSTACK

Bluetooth® Stack

CIS

connected isochronous stream

CSSA

Coding Scheme Selection for Advertising

EDR

enhanced data rate

GATT

generic attribute profile

HCI

host controller interface

Glossary

HFP

hands-free profile

ICS

implementation conformance statement

IOP

interoperable

ISOAL

isochronous adaptation layer

ISOC

isochronous channels

IUT

instrument under test

LL

link layer

LMP

link manager protocol

MIC

message integrity check

MOS

mean opinion score

OS

operating system

PAwR

periodic advertising with responses

PDU

protocol data unit

RPA

resolvable private address

Rx

Reception

SCO

synchronous connection-oriented

SMP

security manager protocol

Glossary

SPP

serial port profile

TCRL

test case reference lists

Tx

transmission

WBS

wide band speech

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